



**DELHI PUBLIC SCHOOL NEWTOWN**  
**SESSION 2023-24**  
**MONDAY TEST**

**CLASS: X**

**SUBJECT: COMPUTER APPLICATIONS**

**TOTAL MARKS: 40**

**DATE: 17.11.23**

**Instructions:**

The Question Paper comprises of Two Parts. Part-I is based on Multiple Choice Questions and Application questions. Part -II is based on Java Programs. The intended marks for questions are given in brackets [ ]. This paper consists of three printed pages.

**PART-I (20 marks)**  
**[Attempt all question]**

**Question 1**

Choose the correct answer from the choices given:

**[6×1=6]**

i) “KANPUR”.substring(3) + “INDIA”.charAt(0) will return:

- a) PURINDIA    b) PURINDI    c) PURI    d) NPURI

ii) Identify the correct syntax to create an object of the class Person.

`class Person`

`{ String name; int age; }`

`Person(String a, int b)`

`{ name=a; age=b; }`

- a) `Person bheem=new Person(“Bheem”, 16);`                      b) `Person bheem=(“Bheem”, 16);`  
b) `New Person bheem=Person(“Bheem”, 16);`                      d) `Person bheem new=(“Bheem”, 16);`

iii) Assertion(A): In java, String is basically an object that represents sequence of char values.

Reason(R): An array of characters does not work same as java string.

- a) Both Assertion (A) and Reason (R) are true and (R) is a correct explanation of (A)  
b) Both Assertion (A) and Reason (R) are true and (R) is not a correct explanation of (A)  
c) Assertion (A) is true and Reason (R) is false  
d) Assertion (A) is false and Reason(R) is true

iv) Given an array 12, 3, 8, 5. If sorting the array in ascending order, what will the array be after two passes of bubble sort?

- a) 12, 8, 5, 3                      b) 3, 8, 12, 5                      c) 3, 5, 8, 12                      d) 12, 3, 5, 8

v) When primitive data type is converted to a corresponding object of its class, it is called

- a) boxing                      b) unboxing                      c) explicit type conversion                      d) implicit type conversion

vi) Which of the following expression is equivalent to `!(c || d)`

- a) `(!c) && (!d)`                      b) `(c || d)`                      c) `!(c && d)`                      d) `(!c) || (!d)`

- a) Predict the output:
- `System.out.println("hi".compareTo("Hi"));`
  - `System.out.println("hi".compareTo("hit"));`
- b) What does `Math.random( )` return?
- c) Name the java class where the following method resides:
- `parseLong("12345")`
  - `toUpperCase('c')`
- d) The following program segment should double the value of each element of the array and then print the new value. Therefore input ( 1, 2, 3, 4, 5 ) should become output (2, 4, 6, 8, 10 ). The program block to do the same has been mixed up. Write the block in the correct order.
- ```

Arr[i] = arr[i] * 2;
for(int i=0;i<arr.length;i++)
{ System.out.println(arr[i]); }
int arr[ ]= {1, 2, 3, 4, 5 };

```
- d) State true or false:
- Relational operators have higher precedence over Arithmetic operators
  - In a pure function, the state of an object is changed.
- e) Anand wants to create a constructor of a class "Employee" which contains instance variables String name, double salary and long mobile\_no, but he is getting errors in the following constructor. Identify the errors and correct them.
- ```

void Employee(String n, int s, double m)
{ name=n; salary=s; m=mobile_no; }

```
- f) Consider the array given below and predict the output: `char ch[ ] = { 'A', 'E', 'I', 'O', 'U' };`
- `System.out.println(ch[1]*2);`
  - `System.out.println(ch[1*2]*1);`

### PART-II (20 marks)

*[Attempt all questions. Each program should be written using Variable descriptions/Mnemonic Codes so that the logic of the program is clearly depicted]*

### Question 3

[10]

An online pizza website company provides a special feature to its users. It allows users to search for toppings to customize their pizza. They have 10 varieties of pizza toppings available, they prompt the user to enter a topping to search. Design a java program to input the varieties of toppings, check if the topping entered by the user exists in an array of toppings or not while customizing. Search the toppings array using the binary search technique and display appropriate message based on whether the topping exists or not.

Example 1 : Input: Enter the topping you want to add to your pizza

Input: olives

Output: This topping is available

**Example 2 : Input: Enter the topping you want to add to your pizza**

**Input: pepperoni**

**Output: This topping is not available**

**Question 4**

**[10]**

**Design a class to overload the method display(.....) as follows:**

- 1. void display(int num) — checks and prints whether the number is a perfect square or not.**  
**A positive integer that is the square of another positive integer is known as a perfect square. It is the result of multiplying two identical integers. For example, 1, 4, 9, 16, 25, etc. are perfect squares because they can be expressed as the square of an integer ( $1^2$ ,  $2^2$ ,  $3^2$ ,  $4^2$ ,  $5^2$  etc.)**
- 2. void display(String str, char ch) — checks and prints if the word *str* contains the letter *ch* or not.**
- 3. void display(String str) — checks and prints the number of special characters present in the word *str*.**

**Write a suitable main( ) method.**

